

PART 4 — WEEK 4 PROJECT SUMMARY

HealthConnect ML Engineering — Week 4 Summary

During Week 4, I focused on designing the initial machine learning system for the HealthConnect Experience Lab. The specific problem identified for the Machine Learning Engineering track is **predicting appointment no-shows** using historical appointment and patient appointment-history data.

I reviewed the HealthConnect appointment dataset and its accompanying data dictionary to understand the available variables, their meanings, data types, and potential modelling dependencies. The dataset contains appointment, scheduling, reminder, historical appointment, distance, waiting-time, and outcome information.

The proposed solution is a supervised classification system that will predict the probability that an upcoming appointment will result in a no-show. The initial workflow will include data ingestion, validation, cleaning, feature engineering, train/test splitting, model training, evaluation, model selection, and eventual model serialization.

Key considerations identified include missing data, class imbalance, potential data leakage, dataset limitations, model overfitting, privacy, fairness, and the need to ensure that only information available before an appointment is used for prediction.

For Week 5, my primary focus will be on **implementing the data-processing and modelling pipeline**, conducting exploratory analysis, preparing the features, establishing a baseline model, and evaluating candidate classification algorithms.